

SINUJU, Korea, Democratic People's Republic of

WMO: 470350

Lat: 40.100N

Lon: 124.383E

Elev: 7

StdP: 101.24

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.5	-13.6	-24.6	0.4	-11.8	-22.4	0.5	-10.8	8.4	-8.7	7.3	-8.1	2.4	0	0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.1	31.3	24.3	29.9	23.6	28.6	23.1	26.2	29.2	25.4	28.0	24.6	26.9	2.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.4	20.6	28.0	24.7	19.7	27.0	23.9	18.8	26.1	81.9	29.5	78.2	28.0	75.0	27.0	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	7.3	6.0	5.1	DB	-18.3	33.8	2.7	1.8	-20.2	35.1	-21.8	36.1	-23.3	37.1	-25.2	38.4
(5)				WB	-18.8	27.7	2.6	1.5	-20.6	28.8	-22.1	29.6	-23.5	30.4	-25.4	31.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	-6.3	-2.7	3.4	10.3	16.4	21.2	24.5	24.8	20.1	12.8	3.8	-4.0	(6)
(7)		DBStd	11.35	4.01	3.93	3.41	3.28	2.62	2.27	1.98	2.08	2.68	3.69	4.86	4.07	(7)
(8)		HDD10.0	1746	504	357	207	35	0	0	0	0	17	192	433		(8)
(9)		HDD18.3	3446	763	590	464	240	71	4	0	14	173	437	692		(9)
(10)		CDD10.0	1900	0	0	1	44	199	335	449	460	302	104	6	0	(10)
(11)		CDD18.3	559	0	0	0	0	11	88	191	202	66	2	0	0	(11)
(12)		CDH23.3	3502	0	0	0	3	81	439	1192	1488	293	6	0	0	(12)
(13)		CDH26.7	841	0	0	0	0	10	91	308	401	31	1	0	0	(13)
(14)	Wind	WSAvg	1.9	2.1	2.0	2.1	2.1	1.9	1.7	1.6	1.7	1.7	1.9	2.1	2.1	(14)
(15)	Precipitation	PrecAvg	968	9	11	18	53	66	112	286	232	93	44	29	12	(15)
(16)		PrecMax	1472	60	41	54	145	144	267	654	508	195	170	61	32	(16)
(17)		PrecMin	670	1	0	2	8	10	21	98	39	17	0	4	1	(17)
(18)		PrecStd	212	11	10	13	35	37	63	144	128	47	37	17	9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.6	9.5	15.8	23.0	28.0	31.1	33.6	33.6	29.1	24.1	16.6	7.5	(19)
(20)			MCWB	1.7	4.1	8.4	14.2	18.0	21.6	26.2	26.0	22.1	17.3	12.5	5.5	(20)
(21)		2%	DB	2.8	6.4	12.4	20.1	25.5	29.0	31.4	31.5	27.5	21.6	14.3	5.1	(21)
(22)			MCWB	0.5	2.3	6.6	12.0	16.6	20.5	24.6	24.9	20.6	16.2	10.9	3.0	(22)
(23)		5%	DB	1.2	4.5	10.4	18.0	23.6	27.4	29.7	30.1	26.3	19.9	12.6	3.4	(23)
(24)			MCWB	-1.2	1.5	5.5	11.0	16.0	20.2	24.1	24.2	20.0	14.9	9.5	1.3	(24)
(25)		10%	DB	-0.3	3.1	8.7	16.2	22.0	25.8	28.2	28.9	25.1	18.5	10.9	1.9	(25)
(26)			MCWB	-2.5	0.5	4.6	10.1	15.4	20.1	23.6	23.8	19.6	14.2	8.3	-0.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	4.8	9.9	15.8	20.1	23.4	27.6	27.7	24.1	19.0	13.9	6.1	(27)
(28)			MCDWB	3.9	7.6	13.7	20.0	25.0	28.1	31.8	31.0	26.9	22.1	15.6	7.1	(28)
(29)		2%	WB	0.7	3.2	7.8	13.6	18.4	22.3	26.2	26.5	22.8	17.5	11.9	3.2	(29)
(30)			MCDWB	2.5	5.5	10.7	17.6	23.0	26.4	29.5	29.2	25.6	20.3	13.8	4.6	(30)
(31)		5%	WB	-0.9	2.0	6.5	12.4	17.3	21.6	25.2	25.7	21.9	16.4	10.2	1.6	(31)
(32)			MCDWB	1.1	4.2	9.2	16.3	21.5	25.2	27.9	28.3	24.7	18.8	12.0	3.1	(32)
(33)		10%	WB	-2.6	0.6	5.2	11.2	16.4	21.0	24.5	25.1	20.9	15.2	8.5	0.0	(33)
(34)			MCDWB	-0.5	2.8	8.1	15.0	20.3	24.2	26.7	27.5	23.7	17.5	10.5	1.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.1	8.5	8.4	9.1	9.1	7.5	5.8	7.1	9.0	9.2	7.6	7.2	(35)
(36)			MCDBR	9.0	9.7	10.7	11.9	12.2	10.8	8.9	9.1	10.1	10.5	9.3	8.1	(36)
(37)			MCWBR	7.2	7.0	6.6	6.0	5.7	4.7	3.8	4.1	4.9	6.1	6.8	6.8	(37)
(38)		5% WB	MCDBR	8.5	8.8	9.6	10.0	10.1	8.3	6.9	6.8	8.0	8.8	8.3	7.7	(38)
(39)			MCWBR	7.2	7.0	6.7	5.9	5.5	4.2	3.6	4.0	4.8	6.1	6.8	7.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.401	0.479	0.540	0.627	0.654	0.736	0.711	0.605	0.531	0.558	0.489	0.415	(40)	
(41)		taud	1.959	1.808	1.730	1.609	1.604	1.522	1.612	1.817	1.915	1.759	1.851	1.949	(41)	
(42)		Ebn at Noon	705	698	705	677	672	618	629	687	709	614	600	649	(42)	
(43)		Edh at Noon	130	175	210	253	260	283	256	204	173	182	145	122	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.41	3.17	4.31	4.98	5.57	4.92	4.16	4.49	4.42	3.30	2.30	2.05	(44)	
(45)		RadStd	0.15	0.22	0.31	0.39	0.45	0.60	0.44	0.52	0.36	0.21	0.24	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-1.80	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page